

CASPER NATRONA COUNTY INTL, WY, USA

WMO: 725690

Lat: 42.898N

Lon: 106.474W

Elev: 5318

StdP: 12.08

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24089

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-7.9	-1.0	-13.8	3.2	-7.2	-7.6	4.5	0.5	39.2	39.6	35.2	36.9	7.0	30	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.7	94.0	59.4	91.4	58.9	88.6	58.3	63.3	82.5	61.9	81.8	60.5	80.9	11.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.8	87.1	66.5	55.7	80.5	66.0	53.7	74.9	65.7	31.2	81.9	30.2	81.4	29.2	80.5	68.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.4	28.2	25.6	DB	-19.5	98.5	6.8	1.8	-24.4	99.9	-28.4	100.9	-32.2	102.0	-37.1	103.3	(4)
(5)			WB	-19.9	65.9	6.6	1.5	-24.7	66.9	-28.6	67.8	-32.3	68.7	-37.1	69.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.2	26.0	27.1	36.1	42.6	52.0	62.8	71.6	69.4	59.5	45.7	34.9	25.3		
	DBStd	18.79	11.27	11.84	10.82	9.09	8.33	7.20	5.13	5.59	8.61	9.98	11.81	12.21		
	HDD50	3607	745	642	440	252	76	5	0	0	22	196	462	767		
	HDD65	7336	1210	1061	896	673	406	121	10	24	202	598	903	1232		
	CDD50	2217	0	0	9	29	139	390	669	602	306	64	9	0		
	CDD65	472	0	0	0	0	4	56	214	161	36	1	0	0		
	CDH74	8433	0	0	1	16	220	1354	3363	2593	842	44	0	0		
	CDH80	3766	0	0	0	0	44	571	1686	1177	284	4	0	0		
(14) Wind	WSAvg	11.7	15.0	13.6	12.6	11.5	10.3	9.9	8.9	9.1	9.8	11.2	13.5	14.9		
(15) Precipitation	PrecAvg	12.3	0.5	0.6	0.9	1.5	2.1	1.4	1.3	0.7	0.9	1.1	0.7	0.6		
	PrecMax	18.9	1.4	1.4	2.3	3.9	6.5	4.7	3.5	2.7	3.4	4.6	2.7	1.4		
	PrecMin	6.6	0.0	0.2	0.1	0.1	0.0	0.1	0.1	0.0	0.1	0.0	0.0	0.1		
	PrecStd	2.8	0.3	0.3	0.5	0.8	1.5	1.1	0.8	0.6	0.8	0.9	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	56.7	69.6	76.4	85.0	95.2	98.1	95.6	90.7	79.3	67.5	53.9		
		MCWB	39.7	42.4	46.2	50.9	55.4	58.0	60.8	59.7	58.9	52.2	45.6	40.2		
	2%	DB	47.1	51.4	63.7	71.1	79.9	91.0	95.0	92.4	86.8	73.7	61.2	48.8		
		MCWB	36.6	39.1	44.0	48.0	54.2	57.9	60.1	58.6	56.3	50.2	43.3	37.9		
	5%	DB	43.7	47.2	58.7	66.4	75.4	87.0	92.4	90.0	83.3	68.9	56.2	45.0		
		MCWB	34.9	37.1	42.0	46.2	53.1	57.1	59.6	58.4	54.8	48.5	41.4	35.6		
	10%	DB	40.7	43.0	53.6	61.3	70.8	82.3	89.5	87.2	79.0	63.7	51.4	41.5		
		MCWB	33.1	34.5	40.0	44.4	51.6	56.4	59.1	58.1	53.9	46.6	39.0	33.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	43.2	47.4	52.4	59.0	63.5	65.7	64.2	61.9	55.1	46.6	41.1		
		MCDB	49.6	55.4	66.2	72.8	78.3	80.9	86.3	83.5	82.9	72.1	63.3	52.1		
	2%	WB	37.6	40.0	44.9	49.2	56.3	61.1	63.9	62.3	59.1	51.6	44.1	38.4		
		MCDB	45.3	50.3	61.1	67.4	73.9	80.5	83.1	82.7	79.1	69.2	59.6	47.7		
	5%	WB	35.4	37.5	42.7	47.1	54.5	59.5	62.4	61.0	57.1	49.5	41.9	36.2		
		MCDB	43.0	46.2	57.3	64.2	71.7	79.3	81.9	81.5	76.8	66.9	54.8	44.4		
	10%	WB	33.3	34.9	40.6	45.1	52.6	58.0	61.3	59.8	55.2	47.3	39.8	33.7		
		MCDB	40.3	42.6	53.1	59.9	68.7	77.3	81.3	80.6	74.1	62.6	50.9	40.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.2	20.4	24.1	25.6	27.3	32.0	33.7	32.7	30.8	26.0	21.7	18.4		
		MCDBR	19.4	22.9	30.6	35.4	35.9	39.7	38.0	36.6	37.0	34.0	27.3	20.0		
		MCWBR	12.2	13.8	15.4	16.9	14.9	14.2	12.1	12.2	14.1	16.1	14.6	12.4		
	5% WB	MCDBR	18.9	22.5	29.3	32.5	32.8	34.5	32.8	31.9	31.6	32.1	26.2	19.7		
		MCWBR	12.5	14.0	15.3	16.2	14.6	13.7	10.9	11.0	12.7	16.0	14.4	12.8		
(40) Clear-Sky Solar Irradiance	taub		0.237	0.247	0.267	0.299	0.317	0.319	0.346	0.341	0.303	0.270	0.250	0.239		
	taud		2.553	2.543	2.527	2.454	2.460	2.469	2.420	2.418	2.521	2.591	2.559	2.540		
	Ebn at Noon		298	311	314	308	303	301	292	290	296	297	289	286		
	Edh at Noon		21	25	29	34	35	34	36	35	29	23	20	19		
(44) All-Sky Solar Radiation	RadAvg		654	919	1328	1667	1974	2360	2318	2012	1612	1078	708	521		
	RadStd		38	58	107	98	162	148	82	55	111	82	58	30		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page